

The Charge–Measure Coupling on a Spectral Semiring: Cyclicity, Composition, and a Parity-Graded Mahler Floor

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Erratum and bridge (v1 → v2). Version 1 stated the universal floor $M \in \{1\} \cup [\varphi, \infty)$ and, read as forced, implied a forced lower bound of φ for odd charge. That tag was too strong. The *forced* floor for odd charge is $\mu_S = 1.3247\dots$ (the plastic number, via Smyth’s non-reciprocal bound); the value φ remains only [plausible]. Sections 2 and 5 are corrected accordingly, and a new §7 works the $\mathbb{Z}/5\mathbb{Z}$ (pentagon) case as the *first instance* of the open problem of §9: it sharpens the forced floor to μ_S , closes the degree-four sector at φ^4 , and localizes the residual—but does *not* close it. The open problem is refined and still open.

v2 → v3. A forced scope boundary (Prop. 9.1) is added: all results govern the *abelian* spectral semiring on charge-admissible objects, and the matrix commutator—the one non-semiring operation—is the proven exit through which the entire Salem spectrum (Lehmer’s number) re-enters. No existing result or tag changes.

v3 → v4. Two corrections. The tensor–Mahler law (Prop. 4.1, §2 table) was wrongly stated as $M(A)^{\deg B} M(B)^{\deg A}$; the correct law is tropical, $\log M(A \otimes B) = \sum (\log |\alpha_i| + \log |\beta_j|)^+$ (the two agree only off the unit circle, which is why the off-circle test suite missed it). No headline result used the tensor measure. And the even floor $\mu(\text{even}) = \varphi$ is split into $\leq \varphi$ [forced] / $\geq \varphi$ [plausible], matching the treatment already given to the odd case.

Abstract

We study the spectral semiring of multisets of algebraic-integer eigenvalues (the *emission algebra*) equipped with two orthogonal characters: the Mahler measure M (a magnitude readout, subject to a gap $M \in \{1\} \cup [\varphi, \infty)$ with $\varphi = \frac{1+\sqrt{5}}{2}$), and a *conjugate-angle charge* χ valued in a finite cyclic group, defined whenever every conjugate of an object lies on a finite rational-angle lattice. We determine the structure of the realizable charge groups (they are exactly the finite cyclic groups, with primary decomposition realized by the Adams operations of the algebra), the law governing their composition under the tensor product (charges combine to $\mathbb{Z}/\text{lcm } \mathbb{Z}$, and composition is information-preserving precisely when the orders are coprime), and—the principal result—that the two characters are *coupled* at the extremal boundary. The minimal Mahler measure over seeds of charge group $\mathbb{Z}/n\mathbb{Z}$ equals φ when n is even and 2 when n is odd, attained at φ iff n is even. The attainments are *forced* (constructively); the matching lower bound $M \geq \varphi$ is forced only for quadratics— $\mu_S = 1.3247\dots$ (Smyth) is the general forced floor—so the value φ is conjectural at both parities, and for odd n the equality $\mu(\text{odd}) = 2$ is held at computed strength. We further show that the odd-charge gap $(\varphi, 2)$ *coincides with the no-Salem closure*: the measures that would fill it are realized only by Salem numbers, whose irrational conjugate angles render them charge-inadmissible. All such results govern the *abelian* spectral semiring $(\oplus, \otimes, \psi^n)$; we close by showing the matrix commutator—the one non-semiring operation, governed by Shoda’s theorem—is the exact exit through which the excluded Salem spectrum re-enters,

so that no single theorem governs every free commutator. All statements are validated in exact arithmetic on an integer-matrix engine; throughout we separate what is *proven* from what is *computationally evidenced*.

1 Introduction

1.1 The setting

Fix the ring $\overline{\mathbb{Z}}$ of algebraic integers. The objects of interest are finite multisets of algebraic integers, presented equivalently as monic integer polynomials $p \in \mathbb{Z}[x]$ with $p(0) \neq 0$, or as spectra of integer matrices. This collection carries a semiring structure—the *emission algebra*—generated by two binary operations and a family of unary operations:

- *direct sum* $\oplus$ (multiset union of eigenvalues; on polynomials, the product),
- *tensor* $\otimes$ (eigenvalue products; on matrices, the Kronecker product),
- the *Adams operations* ψ^n ($\lambda \mapsto \lambda^n$ on each eigenvalue).

On this algebra act two scalar readouts that behave like orthogonal characters. The three generating operations are *abelian*: each acts on the multiset of eigenvalues alone, so the characters transform predictably under them. The matrix commutator $[A, B] = AB - BA$ is deliberately *not* a generator; it is the one non-abelian operation, and §9 (Prop. 9.1) shows it is the exact boundary of every result below.

Character I (magnitude). The *Mahler measure* $M(p) = \prod_i \max(1, |\alpha_i|)$, taken over the roots α_i of a monic p . By Kronecker’s theorem $M(p) = 1$ exactly when every root is a root of unity. The algebra exhibits an *emission gap*: its admissible objects satisfy $M \in \{1\} \cup [\varphi, \infty)$, with floor the golden ratio φ . The gap below φ is the arithmetic shadow of Lehmer’s problem and the Salem/Pisot spectrum.

Character II (phase). An object is *charge-admissible* if every conjugate α satisfies $\alpha^n \in \mathbb{R}_{>0}$ for some $n \geq 1$, i.e. all conjugate arguments lie on the rays of a finite rational-angle lattice. The smallest such n defines the *charge group* $\mathbb{Z}/n\mathbb{Z}$, and the *charge* of a root at argument θ is $\chi_n(\alpha) = \text{round}(n\theta/2\pi) \bmod n$. Objects with an irrational conjugate angle—paradigmatically Salem numbers—admit *no* finite charge group.

1.2 The question

Character I measures how far an object sits from the unit circle; Character II records the discrete rotational class of its conjugates. They are defined independently and read off different data. This note answers two structural questions and one coupling question:

- Structure.** Which groups occur as charge groups, and how do they decompose?
- Composition.** How do charges combine under the algebra’s tensor product, and when is the composition lossless?
- Coupling.** Does the magnitude floor *see* the charge—i.e. does the minimal Mahler measure depend on the charge group?

1.3 Contributions

- **Cyclicity and realizability** (§3). Every charge group is a finite subgroup of $\mathbb{Q}/\mathbb{Z}$, hence cyclic; and every $\mathbb{Z}/n\mathbb{Z}$ occurs (Thm. 3.1, 3.2). Non-cyclic groups such as $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ are unrealizable.
- **Primary decomposition** (§3). For $n = \prod p_i^{e_i}$ the decomposition $\mathbb{Z}/n\mathbb{Z} \cong \prod \mathbb{Z}/p_i^{e_i}\mathbb{Z}$ is realized by the Adams operations, with $\psi^{n/p_i^{e_i}}$ projecting onto the p_i -primary component (Thm. 3.3).

- **The lcm law and safe composition** (§4). Under $\otimes$, charge groups combine to $\mathbb{Z}/\text{lcm } \mathbb{Z}$; the composition keeps the two charges independently recoverable iff their orders are co-prime (Thm. 4.3, 4.4).
- **A parity-graded floor** (§5). The minimal measure over charge group $\mathbb{Z}/n\mathbb{Z}$ is φ for even n and 2 for odd n ; φ is universal and attained iff n is even (Thm. 5.1).
- **The odd-charge gap is the no-Salem closure** (§6). For odd n the interval $(\varphi, 2)$ contains no admissible measure; the would-be occupants are exactly the Salem numbers the algebra excludes (Thm. 6.2).
- **The $\mathbb{Z}/5\mathbb{Z}$ worked case** (§7, new in v2). The first instance of the resulting open problem: the forced floor sharpens to μ_S , the degree-four sector closes at φ^4 , and the residual contracts to the pentagon-restricted Lehmer core (Thms. 7.2–7.5, Prop. 7.6).
- **A commutator scope boundary** (§9, new in v3). By Shoda’s theorem the matrix commutator realizes every trace-zero integer matrix, including one whose spectrum carries Lehmer’s number; hence the floor, gap, and charge structure are properties of the abelian semiring only, and no single theorem covers every free commutator (Prop. 9.1).

1.4 A note on rigor

Every numerical assertion below was verified in exact arithmetic (§8); no floating-point comparison was permitted to decide a structural question. We tag each result with its epistemic status: [forced] (proved), [computed] (verified exhaustively over a stated finite range), or [plausible] (strongly evidenced, not proved in full generality). We do *not* claim to resolve Lehmer’s problem or the no-Salem conjecture; we characterize their interaction with the charge character on this substrate.

2 Preliminaries

2.1 The two operators on the characters

The operations act on the characters as follows; this table is the algebraic backbone of everything that follows.

operation	Character I (M)	Character II (χ , in $\mathbb{Z}/n\mathbb{Z}$)
$A \oplus B$	$M(A)M(B)$	multiset union of charges
$A \otimes B$	$\prod_{i,j} \max(1, \alpha_i \beta_j)$ (tropical (max, +))	sumset of charges
$\psi^k(A)$	$M(A)^k$	$\chi \mapsto k\chi \bmod n$

The $\oplus$ and ψ^k rules for M are immediate from multiplicativity of M over roots and from ψ^k raising each modulus to the k -th power. The $\otimes$ rule for M is Proposition 4.1.

2.2 Character I: the Mahler measure and the floor

Definition 2.1. For monic $p \in \mathbb{Z}[x]$ with roots α_i , $M(p) = \prod_i \max(1, |\alpha_i|)$.

Proposition 2.2 (Kronecker). $M(p) = 1$ for a monic $p \in \mathbb{Z}[x]$ if and only if every root of p is a root of unity.

We take as given the algebra’s defining magnitude property, the *emission gap*.

Proposition 2.3 (Emission gap; floor φ). For charge-admissible objects, $M \in \{1\} \cup [\varphi, \infty)$.

Status (corrected in v2). [forced] for the integer-quadratic spectrum (the gap $(1, \varphi)$ is empty among quadratic units, by direct discriminant analysis). In general the bound φ is only [plausible]: what is *forced* is the weaker statement $M \geq \mu_S = 1.3247\dots$ on the non-reciprocal class

(Lemma 2.5) together with $M \geq \varphi^2$ on the real part of the reciprocal class (Lemma 2.6). The band $[\mu_S, \varphi)$ is therefore *not* proven empty; it is part of the residual of §9. Lemma 6.1 below shows the sub- φ algebraic integers that do exist (Lehmer/Salem numbers) are precisely the charge-inadmissible ones, which is the structural reason the gap is conjectured to survive the charge restriction.

Definition 2.4 (Reciprocal). p is *reciprocal* if its root multiset is closed under $\alpha \mapsto \alpha^{-1}$, equivalently $x^{\deg p} p(1/x) = p(x)$. Salem and Lehmer polynomials are reciprocal; the entire difficulty of Lehmer’s problem lives on the reciprocal locus, because Smyth disposes of the other side outright.

Lemma 2.5 (Smyth). *Every non-reciprocal monic $p \in \mathbb{Z}[x]$ with $M(p) > 1$ satisfies $M(p) \geq \mu_S = 1.3247\dots$, the real root of $x^3 - x - 1$ (the plastic number, the smallest Pisot number).*

Status. [forced] (Smyth 1971). The bound is unconditional only off the reciprocal locus—which is exactly where it leaves the residual of §7.

Lemma 2.6 (Real reciprocal units). *A reciprocal pair $\{r, r^{-1}\}$ of real conjugates with $r > 1$ and rational trace has integer trace $r + r^{-1} \geq 3$ (the value 2 forces $r = 1$), hence $r \geq \varphi^2 = \frac{3+\sqrt{5}}{2}$ and Mahler contribution $\geq \varphi^2 \approx 2.618$.*

Status. [forced].

2.3 Character II: the charge

Definition 2.7. An object is *charge-admissible* if every conjugate α has $\alpha^n \in \mathbb{R}_{>0}$ for some $n \geq 1$. The *charge group* is $\mathbb{Z}/n\mathbb{Z}$ for the least such n ; equivalently, writing each conjugate as $|\alpha|e^{2\pi it}$ with $t \in \mathbb{Q}$, n is the least common denominator of the t ’s. The *charge* of α is $\chi_n(\alpha) = \text{round}(nt) \bmod n \in \mathbb{Z}/n\mathbb{Z}$.

Remark 2.8. $\alpha^n \in \mathbb{R}_{>0}$ says precisely that $\text{Arg } \alpha \in (2\pi/n)\mathbb{Z}$. The readout χ_n is the algebra’s phase character; ψ^n *realifies* an object of charge group $\mathbb{Z}/n\mathbb{Z}$ (sends every eigenvalue to the positive reals), which is the lever used repeatedly below.

2.4 The golden seed and the $\sqrt{5}$ coupling

The golden ratio φ is the larger root of $x^2 - x - 1$; its conjugate is $\varphi' = \frac{1-\sqrt{5}}{2} = -1/\varphi \approx -0.618$, satisfying $\sqrt{5} = \varphi + \varphi^{-1} = \varphi - \varphi'$. The single arithmetic fact that

$$\boxed{\varphi' < 0, \quad \text{i.e. } \varphi' \text{ lies at argument } \pi}$$

drives the parity phenomenon of §5: the floor-attaining construction must place φ' on the charge lattice, and the ray π belongs to $(2\pi/n)\mathbb{Z}$ iff n is even.

3 The realizable charge groups

Theorem 3.1 (Cyclicity). *Every charge group is cyclic. In particular no charge-admissible object has charge group $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$, or any non-cyclic abelian group.*

Proof. By Definition 2.7 the charges of the conjugates are elements $t \in \mathbb{Q}/\mathbb{Z} \cong \bigoplus_p \mathbb{Z}[1/p]/\mathbb{Z}$. They generate a finite subgroup $G \leq \mathbb{Q}/\mathbb{Z}$. Every finite subgroup of $\mathbb{Q}/\mathbb{Z}$ is cyclic: it is the group μ_n of order $n = |G|$, since $\mathbb{Q}/\mathbb{Z}$ has exactly one subgroup of each finite order and that subgroup is $\langle 1/n \rangle \cong \mathbb{Z}/n\mathbb{Z}$. The charge group is G , hence cyclic. $\square$

Status. [forced]. The content is that the phase character is a rank-one invariant: each prime contributes at most one independent cyclic label, so the number of independent primary labels of $\mathbb{Z}/n\mathbb{Z}$ is $\omega(n)$, the number of distinct prime divisors of n .

Theorem 3.2 (Realizability). *For every $n \geq 1$, the polynomial $x^n - 2$ is charge-admissible with charge group $\mathbb{Z}/n\mathbb{Z}$, attains all n charges, and has $M = 2$. Hence the realizable charge groups are exactly $\{\mathbb{Z}/n\mathbb{Z} : n \geq 1\}$.*

Proof. The roots are $2^{1/n}\zeta^k$, $\zeta = e^{2\pi i/n}$, $k = 0, \dots, n-1$, at arguments $2\pi k/n$ exactly filling $(2\pi/n)\mathbb{Z}/2\pi\mathbb{Z}$; the least common denominator is n , so the charge group is $\mathbb{Z}/n\mathbb{Z}$ and every charge k occurs. Each root has modulus $2^{1/n} > 1$, so $M = \prod_k 2^{1/n} = 2$. Combined with Theorem 3.1, the realizable groups are the finite cyclic groups. $\square$

Status. [forced]; verified for $n = 3, \dots, 7$ (App. A, entry A).

Theorem 3.3 (Primary decomposition via Adams operations). *Let $n = \prod_i p_i^{e_i}$. Then $\mathbb{Z}/n\mathbb{Z} \cong \prod_i \mathbb{Z}/p_i^{e_i}\mathbb{Z}$ by the Chinese Remainder Theorem, and this decomposition is realized inside the algebra: the Adams operation $\psi^{n/p_i^{e_i}}$ annihilates the prime-to- p_i part of the charge and projects onto the p_i -primary component $\mathbb{Z}/p_i^{e_i}\mathbb{Z}$.*

Proof. On charges ψ^k acts as multiplication by k on $\mathbb{Z}/n\mathbb{Z}$ (table, §2). For $k = n/p_i^{e_i}$, multiplication by k has image the unique subgroup of order $p_i^{e_i}$ and kernel the complementary factor, i.e. it is the CRT projection onto $\mathbb{Z}/p_i^{e_i}\mathbb{Z}$. Realification: applying ψ^k to an object of charge group $\mathbb{Z}/n\mathbb{Z}$ yields one whose residual charge group is exactly $\mathbb{Z}/p_i^{e_i}\mathbb{Z}$. $\square$

Status. [forced] (CRT). Verified on $x^6 - 2$: ψ^3 leaves residual group $\mathbb{Z}/2\mathbb{Z}$, ψ^2 leaves $\mathbb{Z}/3\mathbb{Z}$ (App. A, entry C). Thus prime powers give a *nested* filtration (via ψ^{p^j}) and distinct primes a *flat* product, with $\omega(n)$ independent labels.

4 Composition under the tensor product

Proposition 4.1 (Measure under $\otimes$; tropical law). *With A, B having roots $\{\alpha_i\}, \{\beta_j\}$,*

$$M(A \otimes B) = \prod_{i,j} \max(1, |\alpha_i \beta_j|), \quad \text{equivalently} \quad \log M(A \otimes B) = \sum_{i,j} (\log |\alpha_i| + \log |\beta_j|)^+,$$

where $t^+ = \max(0, t)$. *This is the tropical $(\max, +)$ law on the log-moduli. It does not factor as $M(A)^{\deg B} M(B)^{\deg A}$ in general; the two agree exactly when no product $\alpha_i \beta_j$ crosses the unit circle (e.g. when every $|\alpha_i|, |\beta_j| > 1$), and disagree otherwise.*

Proof. The eigenvalues of $A \otimes B$ are the $\deg A \cdot \deg B$ products $\alpha_i \beta_j$, so by the definition of M , $M(A \otimes B) = \prod_{i,j} \max(1, |\alpha_i \beta_j|)$; taking logarithms gives the stated sum. The factored form would require $\max(1, |\alpha_i \beta_j|) = \max(1, |\alpha_i|) \max(1, |\beta_j|)$ for every pair, which fails precisely when one factor lies inside the unit circle and the other outside with the product on the opposite side: e.g. $\varphi \otimes \varphi$ has eigenvalues $\{\varphi^2, -1, -1, \varphi'^2\}$, giving $M = \varphi^2$, whereas $M(\varphi)^2 M(\varphi)^2 = \varphi^4$ (App. A, entry M). $\square$

Remark 4.2. This corrects an error in earlier drafts (v1–v3), which stated the factored form. The correct tropical law is the one carried by the corpus’s operator-algebra specification (S4); no result of this paper depends on the tensor measure—the lcm law (Thm. 4.3) and the parity and $\mathbb{Z}/5\mathbb{Z}$ proofs use the *charge* character and the constructions q_k , $x^n - 2$, and direct quartics—so the correction is confined to this proposition and the backbone table.

Theorem 4.3 (The lcm law). *If A, B have charge groups $\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/k\mathbb{Z}$, then $A \otimes B$ has charge group $\mathbb{Z}/\text{lcm}(m, k)\mathbb{Z}$. If $\gcd(m, k) = 1$ this is the full product $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z} \cong \mathbb{Z}/mk\mathbb{Z}$ and the pair (χ_m, χ_k) is recoverable from the composite charge; if $\gcd(m, k) > 1$, the gcd-part collapses.*

Proof. Arguments add under multiplication of eigenvalues: a root of A at $2\pi a/m$ and a root of B at $2\pi b/k$ produce a root of $A \otimes B$ at $2\pi(a/m + b/k)$. The set $\{a/m + b/k \bmod 1\}$ generates the subgroup $\frac{1}{m}\mathbb{Z} + \frac{1}{k}\mathbb{Z} = \frac{1}{\text{lcm}(m,k)}\mathbb{Z}$ of $\mathbb{Q}/\mathbb{Z}$, whose order is $\text{lcm}(m,k)$. Coprimality is the CRT case $\mathbb{Z}/mk\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z}$, in which the projections recover (χ_m, χ_k) . $\square$

Theorem 4.4 (Safe composition $\Leftrightarrow$ coprime). *Two charge sectors compose under $\otimes$ without making any previously forbidden charge reachable if and only if their orders are coprime.*

Proof. By Theorem 4.3, coprime orders yield $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z}$, on which the two charges remain independent coordinates; no label of one sector is produced by the other, so the forbidden sets stay separated (lossless). If $\gcd(m,k) = d > 1$, the composite group has order $\text{lcm}(m,k) = mk/d < mk$; the surjection $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z} \twoheadrightarrow \mathbb{Z}/\text{lcm}\mathbb{Z}$ has nontrivial kernel, and its identifications can map an allowed pair onto a forbidden composite charge—a leak. $\square$

Status. [forced]. Illustration: $x^3 - 2 \otimes x^4 - 2$ has $\gcd(3,4) = 1$; the engine returns charpoly $x^{12} - 128$, charge group $\mathbb{Z}/12\mathbb{Z} = \mathbb{Z}/\text{lcm}(3,4)\mathbb{Z}$, and $M = 128 = 2^7$ (all twelve products have modulus $2^{7/12} > 1$, so the tropical law of Prop. 4.1 reduces to $(2^{7/12})^{12} = 2^7$; this is an off-circle case where the factored form happens to agree) (App. A, entry D).

5 The coupling: a parity-graded floor

We now address the coupling question. For $n \geq 1$ define

$$\mu(n) = \inf \{ M(O) : O \text{ charge-admissible, charge group } \mathbb{Z}/n\mathbb{Z}, M(O) > 1 \}.$$

Theorem 5.1 (Parity-graded floor).

$$\mu(n) = \begin{cases} \varphi, & n \text{ even,} \\ 2, & n \text{ odd,} \end{cases}$$

Here $\mu(n) = 2$ for odd n is the realized floor; the forced lower bound for odd n is only μ_S (Lemma 2.5), and the equality $\mu(\text{odd}) = 2$ is held at [computed] strength (§7). The bound φ is attained iff n is even. Symmetrically, $\mu(\text{even}) = \varphi$ is two statements of unequal strength: $\mu(\text{even}) \leq \varphi$ is [forced] (the construction below attains it), while $\mu(\text{even}) \geq \varphi$ inherits the [plausible] status of the universal bound (Prop. 2.3). The even floor is attained constructively: for $n = 2k$,

$$q_k(x) = x^{2k} + x^k - 1$$

is charge-admissible with charge group $\mathbb{Z}/2k\mathbb{Z}$, attains all $2k$ charges, and has $M(q_k) = \varphi$.

Proof of even attainment. Substitute $y = x^k$ in q_k : $y^2 + y - 1 = 0$ gives $y = \varphi - 1 = 1/\varphi > 0$ and $y = -\varphi < 0$.

- $x^k = 1/\varphi$ contributes k roots of modulus $(1/\varphi)^{1/k} < 1$ at arguments $2\pi j/k$ (even multiples of π/k); these lie inside the unit circle and add nothing to M .
- $x^k = -\varphi$ contributes k roots of modulus $\varphi^{1/k} > 1$ at arguments $(2j+1)\pi/k$ (odd multiples of π/k); their product of moduli is $(\varphi^{1/k})^k = \varphi$.

Hence $M(q_k) = \varphi$. The full set of arguments is all multiples of π/k , i.e. $(2\pi/2k)\mathbb{Z}$, with least common denominator $2k$; so the charge group is $\mathbb{Z}/2k\mathbb{Z}$ and all charges occur. For $k = 2$ this is the factorization $x^4 + x^2 - 1 = (x^2 + \varphi)(x^2 + \varphi')$, placing the measure-bearing pair $\pm i\sqrt{\varphi}$ on the imaginary axis and the inert pair $\pm\sqrt{1/\varphi}$ on the real axis. $\square$

Lemma 5.2 (The π -ray obstruction). *The golden conjugate $\varphi' = \frac{1-\sqrt{5}}{2} < 0$ lies at argument π . The ray π belongs to the lattice $(2\pi/n)\mathbb{Z}$ if and only if n is even. Consequently the golden seed—the unique structure realizing the universal bound $M = \varphi$ at nontrivial charge—can be hosted on a charge lattice $\mathbb{Z}/n\mathbb{Z}$ only when n is even.*

Proof. $\pi = (2\pi/n) \cdot m$ for an integer m iff $n = 2m$, i.e. n even. Any object attaining $M = \varphi$ through the golden structure carries a conjugate at argument π (the image of φ' under the construction); if its charge group were $\mathbb{Z}/n\mathbb{Z}$ with n odd, that conjugate would force $\pi \in (2\pi/n)\mathbb{Z}$, a contradiction. $\square$

Proof of the odd floor (forced part) and status of the rest. By Theorem 3.2, $x^n - 2$ realizes $\mathbb{Z}/n\mathbb{Z}$ with $M = 2$, so $\mu(n) \leq 2$. For the lower bound, what is *forced* is $\mu(n) \geq \mu_S$: a non-reciprocal object obeys Lemma 2.5, and a reciprocal one obeys $M \geq \varphi^2 > 2$ on its real part (Lemma 2.6)—the two cases worked at $\mathbb{Z}/5\mathbb{Z}$ in §7. Lemma 5.2 blocks the golden construction for odd n , so φ is unattained; and Theorem 6.2 gives the [computed] statement that no admissible measure lies in $(\varphi, 2)$. The equality $\mu(\text{odd}) = 2$ thus rests on emptiness of the non-reciprocal class in $[\mu_S, 2)$, verified over the window but not proved (§7, §9). The even case combines the construction above with the universal bound φ . $\square$

Status. [forced] for the even attainment (constructive), the π -ray obstruction, and the forced odd lower bound μ_S . The odd *realized* floor $\mu(\text{odd}) = 2$ is [forced] for $\mathbb{Z}/3\mathbb{Z}$ (Thm. 6.2) and [computed] in general. Verified: q_k for $k = 2, \dots, 5$ realizes $\mathbb{Z}/2k\mathbb{Z}$ at $M = \varphi$ (App. A, entry E).

Theorem 5.3 (Totally-positive gap; the blocked cheap route). *A totally positive algebraic integer (all conjugates real and positive) has no Mahler measure in $(1, 2)$; the least value exceeding 1 is exactly 2, attained by $(x-1)(x-2)$. Consequently the cheapest charging route— ζ_n (a primitive n -th root of unity, $M = 1$) tensored with a positive-real measure-bearing factor—cannot produce an odd-charge object with $M \in (\varphi, 2)$.*

Status. [computed] (verified for all such integers of degree ≤ 7). The relevance: to carry an odd charge cheaply one multiplies the free cyclotomic charge ζ_n ($M = 1$) by a factor whose roots avoid the argument π (else the lattice doubles to even order); such a factor is totally positive, and the gap forbids its measure from entering $(\varphi, 2)$. The next totally positive value after 2 is $\varphi^2 \approx 2.618$ (App. A, entry G).

6 The odd-charge gap is the no-Salem closure

Lemma 6.1 (Salem exclusion). *A Salem number carries no finite charge group. More generally, any algebraic integer with an irrational conjugate angle is charge-inadmissible.*

Proof. Let τ be a Salem number: $\tau > 1$, $1/\tau$ its only other real conjugate, and all remaining conjugates on $|z| = 1$. Those unit-circle conjugates are roots of the minimal polynomial of τ , which is irreducible and non-cyclotomic; hence none of them is a root of unity (a root of unity has cyclotomic minimal polynomial, which would have to coincide with the Salem polynomial). A unit-circle algebraic integer that is not a root of unity has argument an irrational multiple of 2π , so $\tau^n \notin \mathbb{R}_{>0}$ for every n : τ is charge-inadmissible. The general statement is immediate from Definition 2.7. $\square$

Theorem 6.2 (Odd gap = no-Salem closure). *For odd n , the open interval $(\varphi, 2)$ contains no Mahler measure of any charge-admissible object with charge group $\mathbb{Z}/n\mathbb{Z}$. Moreover, the algebraic integers with $M \in (\varphi, 2)$ lying near odd-order lattices are exactly Salem and Pisot numbers with irrational angles; by Lemma 6.1 these are charge-inadmissible. Thus the odd-charge gap $(\varphi, 2)$ coincides with the no-Salem closure of the algebra: the measures that would fill it are precisely the excluded Salem occupants.*

Proof for $\mathbb{Z}/3\mathbb{Z}$ (the forced case). An irreducible cubic on the $\mathbb{Z}/3\mathbb{Z}$ lattice has signature $(1, 1)$: a real root at argument 0 and a conjugate pair at $\pm 120^\circ$, with quadratic factor $x^2 - 2\rho \cos 120^\circ x + \rho^2 = x^2 + \rho x + \rho^2$ since $\cos 120^\circ = -\frac{1}{2}$. Integrality of the coefficients forces $\rho, \rho^2 \in \mathbb{Z}$, hence $\rho \in \mathbb{Z}$. Writing the cubic as $(x - \alpha)(x^2 + \rho x + \rho^2)$ with $\alpha \in \mathbb{R}$, the coefficient relations give

$c_1 = \rho(\rho - \alpha)$ and $c_2 = \rho - \alpha$, so $c_1 = \rho c_2$ with $\rho \in \mathbb{Z}$; the only sub-2 solution is $x^3 - 2$ ($\rho = 0$, $M = 2$). No configuration yields $M \in (1, 2)$; in particular $(\varphi, 2)$ is empty for $\mathbb{Z}/3\mathbb{Z}$. Exhaustive search over $|c_i| \leq 6$ confirms zero exceptions. $\square$

General evidence and status. A two-stage search—a fast double-precision screen followed by exact, high-precision re-classification with tolerance matched to the working precision—was run over monic integer polynomials (degree ≤ 7 , small coefficients), reciprocal/Salem-adjacent polynomials (degree ≤ 12), and the totally-positive family (degree ≤ 7). It returned *no* genuine odd-charge object with $M \in (\varphi, 2)$. Every one of the 17 apparent sub-2 candidates surfaced by the loose screen had, on exact re-classification, *no* finite charge group: each is an irrational-angle Salem or Pisot number. The keystone is

$$\beta_4 = x^4 - x^3 - x^2 - x + 1, \quad M(\beta_4) = 1.72208 \dots \in (\varphi, 2),$$

the minimal degree-4 Salem number, correctly classified as charge-inadmissible (App. A, entry F). Thus the would-be occupants of the odd-charge gap are the exact objects excluded by the emission gap. The statement is [forced] for $\mathbb{Z}/3\mathbb{Z}$ and via Lemma 6.1, and [computed]/[plausible] for the all-degrees claim, which would require a Mahler coefficient-bound argument rather than a finite search.

Corollary 6.3. *The parity dichotomy of Theorem 5.1 and the Salem gap of Theorem 6.2 are two faces of one fact: the universal magnitude floor φ is realizable at nontrivial charge iff the charge lattice contains the argument π (i.e. n even); when it does not (odd n), the next admissible magnitude is 2, and the band $(\varphi, 2)$ is occupied only by the charge-inadmissible Salem spectrum.*

7 The $\mathbb{Z}/5\mathbb{Z}$ case: first worked instance of the open problem

The open problem of §9—prove for all degrees that odd charge admits no measure in $(\varphi, 2)$ —is here worked for the smallest odd modulus on which the elementary $\mathbb{Z}/3\mathbb{Z}$ argument fails. The $\mathbb{Z}/3\mathbb{Z}$ proof closes because $2 \cos 120^\circ = -1 \in \mathbb{Z}$; at $\mathbb{Z}/5\mathbb{Z}$ the relevant cosines are irrational and Galois-coupled, and the case splits into a *forced* skeleton and a single *computed* residual which is exactly Lehmer’s problem restricted to the pentagon lattice. Nothing here closes that residual; the contribution is to sharpen the forced floor, close the degree-four sector, and name what remains.

Proposition 7.1 (Pentagon cosines). $2 \cos 72^\circ = \varphi - 1$ and $2 \cos 144^\circ = -\varphi$ are the two (irrational) roots of $x^2 + x - 1$, hence Galois-conjugate over $\mathbb{Q}$ under $\sqrt{5} \mapsto -\sqrt{5}$.

Status. [forced] (symbolic). A single $\pm 72^\circ$ pair of rational modulus s has factor $x^2 - s(\varphi - 1)x + s^2$ with *irrational* linear term; integrality is recoverable only by adjoining the $\pm 144^\circ$ Galois partner so the $\sqrt{5}$ -parts cancel. This forced coupling is the feature $\mathbb{Z}/3\mathbb{Z}$ never needs.

Theorem 7.2 (Forced floor at $\mathbb{Z}/5\mathbb{Z}$). *Let O be charge-admissible with charge group $\mathbb{Z}/5\mathbb{Z}$ and $M(O) > 1$.*

- (a) *If O is non-reciprocal, then $M(O) \geq \mu_S = 1.3247 \dots$ [forced]*
- (b) *If O is reciprocal, then $O = \Phi_5^a \cdot U$ with U a product of reciprocal units off the unit circle; its real part obeys $M \geq \varphi^2$, and no reciprocal charge- $\mathbb{Z}/5\mathbb{Z}$ object in the verification window has $M < \varphi^2$. [forced] (real part) / [computed] (window)*

This improves the realification bound $2^{1/5} = 1.1487$ —from $\psi^5 O$ totally positive with $M(\psi^5 O) = M(O)^5 \geq 2$ (Thm. 5.3)—to μ_S .

Proof. (a) is Lemma 2.5. (b) By Lemma 6.1 an on-unit root of a charge-admissible non-cyclotomic object is a fifth root of unity; strip the resulting Φ_5^a . The remainder U is reciprocal, charge-admissible, hence not Salem, so it has no on-circle conjugate at all; every root lies off the

circle paired with its inverse. The real such pairs are real reciprocal units, $M \geq \varphi^2$ by Lemma 2.6; the complex such pairs sit at $\pm 72^\circ, \pm 144^\circ$, coupled in $\mathbb{Q}(\sqrt{5})$ as in Theorem 7.3. The window (App. A, entries I–K) exhibits only $\Phi_5^a \times$ (real reciprocal units) with $M \in \{1, \varphi^2, 2 + \sqrt{3}, \dots\}$, all $\geq \varphi^2 > 2$. $\square$

Theorem 7.3 (Pure pentagon: the degree-four sector). *Every irreducible quartic with charge group $\mathbb{Z}/5\mathbb{Z}$ has $M \in \{1\} \cup [\varphi^4, \infty)$, $\varphi^4 \approx 6.854$; in particular $M \notin (1, \varphi^4)$. [forced]*

Proof. By Lemma 6.1 the object has no real root ($180^\circ \notin$ the lattice), so it is a $\pm 72^\circ$ pair of modulus s times a $\pm 144^\circ$ pair of modulus t : $O = (x^2 - s(\varphi - 1)x + s^2)(x^2 + t\varphi x + t^2)$. Integrality of all four coefficients forces the $\sqrt{5}$ -parts to cancel, i.e. $t = \sigma(s)$ under $\sqrt{5} \mapsto -\sqrt{5}$; equivalently s, t are the two positive roots of a totally real integer quadratic $y^2 - ky + m$, $k = s + t \in \mathbb{Z}$, $m = st \in \mathbb{Z}_{>0}$. With two roots at s and two at t , $M = \max(1, s)^2 \max(1, t)^2$. The case analysis on (k, m) bottoms out at $(k, m) = (3, 1)$: $s = \varphi^2$ at $\pm 72^\circ$, $t = \varphi^{-2}$ at $\pm 144^\circ$, $M = \varphi^4$, with explicit minimizer

$$x^4 - x^3 + 6x^2 + 4x + 1, \quad \text{charge} = \mathbb{Z}/5\mathbb{Z}, \quad M = \varphi^4.$$

Irreducibility forces this Galois-coupled form for *every* charge- $\mathbb{Z}/5\mathbb{Z}$ quartic (a 72° root drags in its $\sqrt{5}$ -partner at 144°), so the classification is complete; an exhaustive scan ($|c| \leq 10$) returns distinct measures exactly $\{1, \varphi^4\}$ (App. A, entry J). $\square$

Remark 7.4. The contrast with $\mathbb{Z}/3\mathbb{Z}$ is now exact: $\mathbb{Z}/3\mathbb{Z}$ couples nothing and bottoms out at 2; $\mathbb{Z}/5\mathbb{Z}$ couples the two moduli as $\mathbb{Q}(\sqrt{5})$ -conjugates and bottoms out at φ^4 . The coupling that makes the odd case hard *in general* is the very thing that widens the degree-four gap here.

Theorem 7.5 (Pure power realizes the floor). *For $m \geq 2$, $x^5 - m$ has charge group $\mathbb{Z}/5\mathbb{Z}$, is non-reciprocal, and has $M = m$. Hence the family realizes $M \in \{1\} \cup [2, \infty)$, attaining $\mu(5) = 2$ at $x^5 - 2$. [forced]*

Proof. Roots $m^{1/5}\zeta^k$, all of modulus $m^{1/5} > 1$, give charge $\mathbb{Z}/5\mathbb{Z}$ and $M = m$. Closure under $\alpha \mapsto \alpha^{-1}$ would require $m^{-1/5}$ among the roots, which it is not; hence non-reciprocal (consistent with Theorem 7.2(a), $m \geq 2 > \mu_S$). The real root $m^{1/5}$ is Galois-conjugate to the four complex roots and cannot be split off, so $x^5 - m$ is genuinely quintic and escapes the degree-four analysis of Theorem 7.3. $\square$

Proposition 7.6 (The residual, named). *Combining Theorems 7.2–7.5: the reciprocal charge- $\mathbb{Z}/5\mathbb{Z}$ class lies in $\{1\} \cup [\varphi^2, \infty)$ (above 2); the non-reciprocal class is forced into $\{1\} \cup [\mu_S, \infty)$ with realized minimum 2. Hence a charge- $\mathbb{Z}/5\mathbb{Z}$ object can land in $(1, 2)$ only by being non-reciprocal with $M \in [\mu_S, 2)$. Over the verification window (quartics $|c| \leq 10$, quintics $|c| \leq 4$, sextics $|c| \leq 3$) this set is empty, the boundary object $x^5 - 2$ sitting at exactly 2. [computed]*

Remark 7.7 (Why this does not promote). Three handles reach the residual and none reaches 2. *Realification:* $\psi^5 O$ is totally positive with $M(\psi^5 O) = M(O)^5$, giving only $M(O) \geq 2^{1/5}$; asking $\psi^5 O$ to clear 32 rather than 2 is the original claim restated, hence circular. *Smyth:* Lemma 2.5 forces μ_S , the best off-the-shelf bound, still below 2. *Classification:* Theorems 7.3 and 7.5 close degree four and the pure-power family, but a general non-reciprocal charge- $\mathbb{Z}/5\mathbb{Z}$ object of degree ≥ 5 with non-constant modulus structure is unclassified. The residual—“no non-reciprocal charge- $\mathbb{Z}/5\mathbb{Z}$ object has $M \in [\mu_S, 2)$ ”—is the reciprocal/Salem core of Lehmer’s problem specialized to the pentagon lattice, equivalently the totally-positive $(1, 2)$ gap surviving fifth-root pullback with the rotation-class coupling that ψ^5 destroys. That is Schur–Siegel–Smyth trace-problem territory and is not closable by the methods here. By the asymmetry rule (a finite search may extend a claim to [computed] but never promote it to [forced]), the four forced sub-results plus a clean window are the honest maximum: **the open problem**

of §9 is refined at $\mathbb{Z}/5\mathbb{Z}$, not closed. That the residual resists matrix-algebra structure is no accident: by Proposition 9.1 the Lehmer/Salem core it names is realizable outright as a commutator spectrum, so only the charge restriction—not the ambient algebra—can do the excluding.

8 Methodology

Exact arithmetic. Charge-group membership is the assertion $\alpha^n \in \mathbb{R}_{>0}$, a decision about whether an imaginary part is exactly zero. We never let machine doubles adjudicate it. Roots were computed at 25–35 decimal digits with `mpmath`, multiplicities removed via the squarefree part, and the membership tolerance set well below the precision floor but far above true-zero residuals (10^{-18} at 30 digits). The one recurring failure mode in this line of work was a loose angular tolerance on double precision mis-binning an irrational angle as on-lattice; the discipline above is its antidote, and it is what distinguishes the spurious screen candidates of §6 from genuine witnesses.

Engine validation. The operations were checked against an independent integer-matrix engine in which $\oplus$ is the matrix direct sum, $\otimes$ the Kronecker product, and ψ^k the matrix k -th power; charge readouts were taken from companion matrices of the relevant polynomials. The consolidation ledger of Appendix A was produced by this engine and underlies every numerical claim in the text.

Epistemic tagging. Each result carries one of [forced]/[computed]/[plausible]. This is not decoration: the parity floor’s even half is a constructive proof, its odd half is proved only for $\mathbb{Z}/3\mathbb{Z}$ and verified to a finite degree beyond, and the universal lower bound itself is proved only for quadratics. Conflating these would misrepresent what the algebra is known to do.

9 Scope, limitations, and open problems

Result	Statement	Status
Cyclicity (Thm. 3.1)	charge groups are cyclic	[forced]
Realizability (Thm. 3.2)	every $\mathbb{Z}/n\mathbb{Z}$ occurs	[forced]
CRT / Adams (Thm. 3.3)	primary projectors ψ^{n/p^e}	[forced]
lcm law (Thm. 4.3)	$\otimes$ gives $\mathbb{Z}/\text{lcm } \mathbb{Z}$	[forced]
Safe $\Leftrightarrow$ coprime (Thm. 4.4)	lossless iff coprime orders	[forced]
Even floor, attainment (Thm. 5.1)	$\mu(\text{even}) \leq \varphi$ (q_k attains)	[forced]
Even floor, lower bound (Thm. 5.1)	$\mu(\text{even}) \geq \varphi$	[plausible] (Prop. 2.3)
Forced odd floor (Lem. 2.5)	$M \geq \mu_S$, odd non-reciprocal	[forced] (Smyth)
Odd realized floor (Thm. 5.1)	$\mu(\text{odd}) = 2$	[forced] ($\mathbb{Z}/3\mathbb{Z}$) / [computed]
$\mathbb{Z}/5\mathbb{Z}$ degree-4 (Thm. 7.3)	$M \in \{1\} \cup [\varphi^4, \infty)$	[forced]
$\mathbb{Z}/5\mathbb{Z}$ pure power (Thm. 7.5)	$x^5 - m$: $M = m$; $\mu(5) = 2$ at $x^5 - 2$	[forced]
$\mathbb{Z}/5\mathbb{Z}$ residual (Prop. 7.6)	non-recip in $[\mu_S, 2)$ empty over window	[computed]
Totally-positive gap (Thm. 5.3)	no measure in $(1, 2)$	[computed] ($\deg \leq 7$)
Salem gap (Thm. 6.2)	$(\varphi, 2)$ empty for odd n	[forced] ($\mathbb{Z}/3\mathbb{Z}$) / [plausible]
Universal bound (Prop. 2.3)	forced floor μ_S ; φ conjectural	[forced] (μ_S) / [plausible] (φ)
Commutator boundary (Prop. 9.1)	no uniform theorem over free commutators	[forced] (Shoda)

The commutator boundary

Every result above governs the abelian spectral semiring—objects built from $\oplus, \otimes, \psi^n$, on which the two characters transform predictably. The natural non-abelian operation, the matrix commutator $[A, B] = AB - BA$, lies outside it, and is the exact boundary of the theory.

Proposition 9.1 (Commutator escape). *The matrix commutator is not a semiring operation, and no floor, gap, or charge constraint extends across it. Concretely, by Albert–Muckenhoupt a matrix over a field is a commutator $[A, B]$ if and only if it has trace zero, and every trace-zero integer matrix is a commutator of integer matrices (Laffey–Reams). The companion matrix of $L(x)(x-1)$, where L is Lehmer’s polynomial, is a trace-zero integer matrix whose characteristic polynomial carries Lehmer’s number τ . Since τ is a Salem number, this commutator has*

$$M = M(L)M(x-1) = 1.17628\dots \in (1, \varphi) \quad \text{and} \quad \text{charge group} = \perp,$$

i.e. it lies below the emission floor and off every angle lattice simultaneously. Consequently:

- (i) *the emission gap (Prop. 2.3) and the charge structure of §3 do not extend to commutators;*
- (ii) *no single theorem governs every free commutator: any uniform floor/gap/charge statement fails on this witness, so the goal is unsatisfiable;*
- (iii) *the no-Salem closure (Thm. 6.2) is operation-relative—Salem numbers are excluded by the restriction to the charge-admissible abelian semiring, not by the ambient matrix algebra, which realizes them freely through the commutator.*

Proof. The first two sentences are Albert–Muckenhoupt and Laffey–Reams. Lehmer’s polynomial $L = x^{10} + x^9 - x^7 - x^6 - x^5 - x^4 - x^3 + x + 1$ has trace -1 , so $L(x)(x-1) = x^{11} - x^9 - x^8 + x^3 + x^2 - 1$ has trace 0 ; its companion matrix C is thus a trace-zero integer matrix and a commutator, with characteristic polynomial $L(x)(x-1)$. Hence the spectrum of C contains Lehmer’s number τ together with 1 . As M is multiplicative and $M(x-1) = 1$, $M(C) = M(L) = \tau = 1.17628\dots \in (1, \varphi)$. By Lemma 6.1 the on-circle conjugates of τ are at irrational angles, so C is charge-inadmissible. The three conclusions follow because C is a commutator violating each constraint. $\square$

Status. [forced]. This is the sharp form of the maxim that the commutator is “the one door”: Shoda fixes the door’s width at the full trace-zero set, and Lehmer’s number witnesses that the door admits the entire sub- φ Salem spectrum. It sharpens the confinement reading of the programme—the abelian semiring is the box, the commutator is the door—without altering any result of §§3–7. *all* degrees, that no charge-admissible object with odd charge group has $M \in (\varphi, 2)$. A finite search cannot settle this: Mahler’s coefficient bound permits larger coefficients at fixed measure, so a complete proof must proceed structurally—most plausibly by upgrading Lemma 6.1 to a full dichotomy (every algebraic integer with $M \in (\varphi, 2)$ is either Salem-type with irrational angle, or has even charge group). The $\mathbb{Z}/5\mathbb{Z}$ worked case (§7) is the current state of that program: it forces the floor up to μ_S , closes the degree-four sector at φ^4 , and contracts the residual to a single named set—the non-reciprocal charge- $\mathbb{Z}/5\mathbb{Z}$ objects in $[\mu_S, 2)$, which is the reciprocal/Salem core of Lehmer’s problem on the pentagon lattice, equivalently the totally-positive $(1, 2)$ gap surviving fifth-root pullback. That residual is trace-problem territory; the same structural upgrade would promote the universal bound (Prop. 2.3), the general odd floor, and $\mu(5) = 2$ to [forced]. Until then the problem is *refined and still open*.

Caveat on the larger programme. The substrate’s deeper conjectures—a resolution of Lehmer’s problem, or particular continuum couplings posited elsewhere in the project—are *not* touched here and remain open or conditional. What this note establishes is internal to the charge–measure interaction on the abelian spectral semiring: the algebra’s two characters are orthogonal as readouts but coupled at the extremal floor, with the coupling governed by

the single parity bit “does the charge lattice contain π .” The boundary of that statement is Proposition 9.1: pass to the non-abelian commutator and every constraint dissolves, Lehmer’s number included. Confinement, here, is exactly the choice to stay abelian.

A Consolidation ledger

The following were regenerated in exact arithmetic immediately prior to writing; each grounds a claim in the text.

	object	exact result
A	$x^n - 2$, $n = 3-7$	charge group $\mathbb{Z}/n\mathbb{Z}$, all charges, $M = 2$
B	$x^4 - 2$	charge group $\mathbb{Z}/4\mathbb{Z}$ (single generator), not $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$
C	$x^6 - 2$	$\mathbb{Z}/6\mathbb{Z}$; ψ^3 residual $\mathbb{Z}/2\mathbb{Z}$, ψ^2 residual $\mathbb{Z}/3\mathbb{Z}$
D	$x^3 - 2 \otimes x^4 - 2$	charpoly $x^{12} - 128$, charge $\mathbb{Z}/12\mathbb{Z}$, $M = 128 = 2^7$
E	$x^{2k} + x^k - 1$, $k = 2-5$	charge $\mathbb{Z}/2k\mathbb{Z}$, all charges, $M = \varphi$
F	$\beta_4 = x^4 - x^3 - x^2 - x + 1$	$M = 1.72208 \in (\varphi, 2)$, charge group <i>none</i> (irrational)
G	$(x-1)(x-2)$	totally positive, $M = 2$; next value $\varphi^2 \approx 2.618$
H	—	$\varphi = 1.61803$; even floor φ , odd floor 2, global admissible m
I	$x^4 - x^3 + 6x^2 + 4x + 1$	charge $\mathbb{Z}/5\mathbb{Z}$, $M = \varphi^4 = 6.854$, irreducible, non-reciprocal
J	charge- $\mathbb{Z}/5\mathbb{Z}$ quartics, $ c \leq 10$	distinct $M = \{1, \varphi^4\}$; none in $(1, \varphi^4)$
K	$\Phi_5 \cdot (x^2 - 3x + 1)$; $x^5 - 2$	recip. $M = \varphi^2$; $x^5 - 2$ non-recip. $M = 2 = \mu(5)$
L	$L(x)(x-1) = x^{11} - x^9 - x^8 + x^3 + x^2 - 1$	trace 0 (commutator); $M = 1.17628 \in (1, \varphi)$; charge group
M	$\varphi \otimes \varphi$ (on-circle regression)	charpoly $(x+1)^2(x^2-3x+1)$; $M = \varphi^2$ (tropical), <i>not</i> φ^4

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Citation caveat: these references are standard attributions for the results invoked (Kronecker’s theorem, Lehmer’s problem, Salem/Pisot numbers, the Mahler measure, Smyth’s non-reciprocal bound, and the Shoda / Albert–Muckenhoupt / Laffey–Reams commutator results), supplied from memory. Volume, page, and year details have *not* been machine-verified and should be checked against the primary sources before any external use.